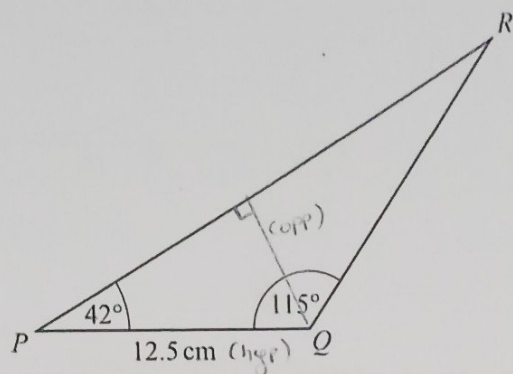


8

12



NOT TO
SCALE

The diagram shows triangle PQR .

Calculate the shortest distance from Q to PR .

$$\sin \theta = \frac{\text{opp}}{\text{hyp}} \quad 8.96 \text{ cm} = \text{opp}$$

$$\sin 42^\circ = \frac{\text{opp}}{12.5}$$

$$\sin 42^\circ \times 12.5 = \text{opp}$$

$$\dots\dots\dots 8.96 \dots\dots\dots \text{ cm [3]}$$

13 Make x the subject of this formula.

$$A = w^2 + 5x^2$$

$$\frac{A - w^2}{5} = \frac{5x^2}{5}$$

$$\frac{A - w^2}{5} = x^2$$

$$\pm \sqrt{\frac{A - w^2}{5}} = x$$

$$x = \pm \sqrt{\frac{A - w^2}{5}} \quad [3]$$